

Mereck McGowan

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Professional Summary

Software Engineer & Machine Learning Researcher with hands-on experience building production web applications and data-driven systems using React, Python, AWS, Docker, and SQL. Founder of a software consultancy delivering full-stack solutions to real clients. Experienced in explainable machine learning, AI, automation, and applied data analysis across academic research and enterprise environments. Seeking **software engineering, full-stack development, and machine learning roles**.

Technical Skills

Languages: Python, Java, JavaScript, C++, C, SQL, R, Racket, Assembly

Web & Data: Backend - REST APIs, SQL databases, NoSQL databases, data pipelines **Frontend:** React.js, Next.js HTML, CSS

Machine Learning: PyTorch, neural networks, explainable ML, reinforcement learning

Cloud & DevOps: AWS EC2, Docker, Nginx, Linux, Git, GitHub, CI/CD

Education

Bachelor of Science, Computer Science - Business Minor | Berry College - Mt. Berry, GA

05/2026

Data Structures & Algorithms, Designing Algorithms, Machine Learning, Artificial Intelligence, Building Programming Languages, Computing Systems, Web Development, Data Analysis

Internship

Data Analyst / Automation Developer | Mohawk Industries - Calhoun, GA

05/2025 - 08/2025

- Designed and implemented automated pay audit system using **Python and SQL**, reducing review time and improving accuracy
- Built an asset utilization analysis system to evaluate logistics truck performance and efficiency

Experience

Software Developer | Owner - Entrepreneur - Asheville, NC

05/2024 - Present

- Built and deployed full-stack React applications for clients, using Dockerized AWS infrastructure with Nginx and SSL for production-grade reliability
- Maintained production servers and client websites, handling updates, uptime, and performance monitoring

Machine Learning Researcher | Berry College - Rome, GA

05/2025 - 05/2026

- Developed fully explainable neural network models using NN-kNN, achieving comparable or improved accuracy over traditional black-box models
- Research paper accepted into the **2026 International Joint Conference on Artificial Intelligence**

Teaching Assistant | Berry College - Rome, GA

08/2025 - 05/2026

- Supported instruction for Object-Oriented Programming and Program Design courses; assisted students with Java and abstraction concepts, graded assignments, and developed instructional materials

Math and Computer Science Ambassador | Berry College - Rome, GA

09/2025 - 05/2026

- Created and hosted programs, advised students, and promoted the Math and Computer Science programs

Employer Engagement Specialist | Berry College - Rome, GA

05/2024 - 05/2026

- Managed and analyzed job and applicant data using **Python, Excel, and R**
- Built analytical models from Handshake and JobX data to support hiring and reporting insights
- Collaborated cross-functionally with employer engagement and career development teams

Projects

Evala | Programming Language for Automated Testing | github.com/mereck600/Evala

- Designed and implemented a Java-based programming language to automatically grade code and generate robust test cases
- Built a custom parser, interpreter, and test generation framework inspired by Java-like syntax

DelphiShell | Local LLM Command Shell | github.com/mereck600/MereckOS

- Designed and built a C based shell with a fine-tuned LLM (GPT 2.0) based natural language to Bash translator
- Designed custom LLM training pipelines, python-based model servers, with Makefile build pipelines

Explainable Neural Network (NN-kNN + RL)

- Designed fully explainable neural network models combining NN-kNN and reinforcement learning
- Achieved accuracy comparable to or exceeding traditional neural networks while maintaining interpretability
- Implemented in Python with PyTorch and custom training pipelines

Involvement

Berry College Computer Science Club | *President* 2024- 2026

Berry College Men's Lacrosse | *Player* 2022 - 2024